Answer one of the following questions in an essay about 2500 words long. Make sure to engage with the relevant readings, and acknowledge any help you receive.

Axelrod

Axelrod's ecological simulations suggest that cooperative strategies like Tit for Tat can thrive in a population of defectors, provided certain conditions are met. But the model assumes perfect implementation. In the original verison, players always execute the move they intend. When noise is introduced, TFT's core strength (immediate retaliation) can become a liability. Does the failure of TFT in noisy environments undermine the broader lesson Axelrod draws about the evolution of cooperation, or does it refine it? In your answer, discuss at least one alternative strategy (e.g., Generous TFT, Contrite TFT, Pavlov) and explain what its success in noisy settings tells us about the conditions under which cooperation can be sustained.

Spence

The Spence signaling model explains the college wage premium without assuming that college teaches useful skills. The model requires that college is more costly for low-productivity workers than for high-productivity ones. Evaluate this assumption. Is there good reason to think the cost-type correlation holds in practice? Draw on at least two of the following in your answer: the sheepskin effect, the age profile of the wage premium, cross-country evidence on degree length (e.g., the Bologna reforms and the Los Andes experiment), and the "permanent student" problem discussed in class.

Zollman

The Zollman effect shows that less connected scientific networks can reach the correct consensus more often than highly connected ones. Rosenstock, Bruner, and O'Connor argue that this effect is confined to a small region of the parameter space and largely disappears when the evidence is even moderately informative. Does this robustness analysis weaken the philosophical interest of the Zollman effect, or does the effect remain valuable because that part of the parameter space is important, or for some other reason? In your answer, explain the effect, why Rosenstock et al think it is limited, and why these limits matter for how we use the model.

Credit

Merton argued that eminent scientists receive disproportionate credit for their work, while lesser-known scientists receive too little. Strevens responds that this pattern may not be unfair, and might even benefit science. Heesen and Rubin argue that it can be driven by luck and network position rather than quality. Who has the better of this argument? Is the Matthew effect, on balance, good or bad for scientific communities? In your answer, consider both the potential benefits (e.g., the symmetry-breaking argument for division of labour) and the potential harms (e.g., the Matilda effect, runaway reputation dynamics).